

Energy, Emissions and AI

Posted by u/cognitive_courier • 0 points • 2 comments

Hey! I write a newsletter covering the AI and tech world, and just wrote an article about the intersection of energy, emissions and AI that I wanted to share with you.

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Artificial intelligence (AI) has the potential to revolutionize the energy sector, but it also brings significant challenges. As AI systems and data centers expand, their electricity consumption is straining power grids, potentially increasing greenhouse gas emissions. Currently, AI computing accounts for less than 1 percent of global emissions, but this could grow substantially depending on the pace of AI advancement and energy choices. Companies like Google aim for net-zero emissions by 2030, yet their emissions have surged, driven by AI and data center expansions.

In the medium term, AI can enhance energy production and efficiency, contributing to cleaner energy use. AI systems can optimize renewable energy generation, predict weather patterns, and manage energy storage, helping to balance supply and demand. Additionally, AI can accelerate research by analyzing vast datasets to identify promising materials for batteries, carbon capture, and other technologies. This could lead to significant advancements in energy production, such as breakthroughs in nuclear fusion and hydrogen energy, while reducing costs and emissions.

Looking ahead, AI's long-term impact on energy is uncertain but potentially transformative. Advanced AI systems could simulate entire societies to develop climate-friendly policies with broad acceptance. However, as AI grows more powerful, it will require significantly more energy, driving demand for cleaner and more efficient energy sources. The Gulf countries, with their resources and focus on AI, could benefit greatly by integrating AI with human expertise to address key challenges. Ensuring the availability of reliable and transparent data will be crucial for maximizing AI's potential in transforming the energy landscape.

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I would be curious to hear from others - what are some specific ways AI can optimize renewable energy generation and storage? Do you think tech firms such as Google will ever achieve their emissions goals? What are the potential risks and benefits of AI systems simulating entire societies to develop climate-friendly policies?

If you enjoyed this you can see more of my work at [The Cognitive Courier]

(<https://cognitivecourier.com>)'.

Comments

u/RoundLife795 • 2 points • 2024-07-09 09:56 UTC

Good stuff, u/cognitive_courier

Thanks for sharing it.

Your article aligns with the ideas of great thought leaders (e.g., this interview with [this interview to Bill Gates on the subject](<https://www.theguardian.com/technology/article/2024/jun/27/ai-bill-gates-climate-targets-datacentres-energy>)).

There are initiatives and lines of research to develop neural networks that are more energy-efficient than our current ANNs.

[https://www.youtube.com/watch?v=WbLQqPw_n88](https://www.youtube.com/watch?v=WbLQqPw_n88)

[https://www.ted.com/talks/frances_s_chance_are_insect_brains_the_secret_to_great_ai/comments/transcript?subtitle=en]

(https://www.ted.com/talks/frances_s_chance_are_insect_brains_the_secret_to_great_ai/comments/transcript?subtitle=en)

u/AutoModerator • 1 points • 2024-07-08 10:05 UTC

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Question Discussion Guidelines

Please use the following guidelines in current and future posts:

- * Post must be greater than 100 characters - the more detail, the better.
- * Your question might already have been answered. Use the search feature if no one is engaging in your post.
- * AI is going to take our jobs - its been asked a lot!
- * Discussion regarding positives and negatives about AI are allowed and encouraged. Just be respectful.
- * Please provide links to back up your arguments.
- * No stupid questions, unless its about AI being the beast who brings the end-times. It's not.

Thanks - please let mods know if you have any questions / comments / etc

I am a bot, and this action was performed automatically. Please [contact the moderators of this subreddit] (/message/compose/?to=r/ArtificialIntelligence) if you have any questions or concerns.